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TREND-BASED MOCK PAPER

GATE Metallurgical Engineering

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Metallurgical Engineering
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

10% of 500:

- (A) 5
- (B) 50
- (C) 500
- (D) 0.5

Section B: Metallurgical Engineering

Q2. [1-MCQ] [1 mark]

Rank of 3x3 zero matrix:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q3. [1-NAT] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

(Numerical Answer Type — Enter the numerical value)

Q4. [1-MCQ] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q5. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q6. [1-NAT] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q7. [1-MCQ] [1 mark]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q8. [1-MCQ] [1 mark]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
- (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q9. [1-MCQ] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q10. [1-MCQ] [1 mark]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q11. [1-NAT] [1 mark]

$P(|Z| < 1)$ for $Z \sim N(0,1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q12. [1-MCQ] [1 mark]

For the ODE $\frac{dy}{dx} = \frac{y}{x} + \sqrt{\frac{y^2}{x^2}}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q13. [1-MCQ] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q14. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q15. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q16. [2-NAT] [2 marks]

Number of spanning trees in complete graph K_4 :

(Numerical Answer Type — Enter the numerical value)

Q17. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q18. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q19. [2-MSQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q20. [2-MCQ] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$

(D) (1,2,3)

Q21. [2-MSQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q22. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q23. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C * x$
- (B) $y = C/x$
- (C) $y = C * x^2$
- (D) $x^2 + y^2 = C$

Q24. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q25. [2-NAT] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q26. [2-MCQ] [2 marks]

The Laplace transform of $t * e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q27. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q28. [2-NAT] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q29. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q30. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q31. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)} \cdot u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q32. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q33. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q34. [1-MCQ] [1 mark]

BCC coordination number:

- (A) 4
- (B) 6
- (C) 8
- (D) 12

Q35. [1-MCQ] [1 mark]

Eutectic reaction:

- (A) Liquid $\rightarrow$ α + β
- (B) $\alpha \rightarrow$ β + γ
- (C) Liquid $\rightarrow$ α

(D) $\alpha + \beta \rightarrow \gamma$

Q36. [1-MCQ] [1 mark]

BCC structure coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q37. [1-MCQ] [1 mark]

Phase diagram lever rule finds:

- (A) Temperature
- (B) Phase fraction
- (C) Composition
- (D) Density

Q38. [2-MCQ] [2 marks]

Eutectic reaction:

- (A) $\text{Liquid} \rightarrow \alpha + \beta$
- (B) $\alpha \rightarrow \beta + \gamma$
- (C) $\text{Liquid} \rightarrow \alpha$
- (D) $\alpha + \beta \rightarrow \gamma$

Q39. [2-MCQ] [2 marks]

BCC structure coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q40. [2-MCQ] [2 marks]

Eutectic reaction:

- (A) $\text{Liquid} \rightarrow \alpha + \beta$
- (B) $\alpha \rightarrow \beta + \gamma$
- (C) $\text{Liquid} \rightarrow \alpha$
- (D) $\alpha + \beta \rightarrow \gamma$

Q41. [2-NAT] [2 marks]

BCC structure coordination number:

(Numerical Answer Type — Enter the numerical value)

Q42. [2-MCQ] [2 marks]

Phase diagram lever rule finds:

- (A) Temperature
- (B) Phase fraction
- (C) Composition
- (D) Density

Q43. [2-MSQ] [2 marks]

Phase diagram lever rule finds:

- (A) Temperature
- (B) Phase fraction
- (C) Composition
- (D) Density

Q44. [2-MCQ] [2 marks]

Phase diagram lever rule finds:

- (A) Temperature
- (B) Phase fraction
- (C) Composition
- (D) Density

Q45. [1-MCQ] [1 mark]

Brass is:

- (A) Cu-Zn
- (B) Cu-Sn
- (C) Cu-Ni
- (D) Cu-Al

Q46. [1-NAT] [1 mark]

BCC coordination number:

(Numerical Answer Type — Enter the numerical value)

Q47. [1-NAT] [1 mark]

Steel hardening process:

(Numerical Answer Type — Enter the numerical value)

Q48. [1-MCQ] [1 mark]

FCC coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q49. [2-MCQ] [2 marks]

Steel hardening process:

- (A) Annealing
- (B) Quenching
- (C) Normalizing
- (D) Tempering

Q50. [2-MCQ] [2 marks]

FCC coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q51. [2-NAT] [2 marks]

BCC coordination number:

(Numerical Answer Type — Enter the numerical value)

Q52. [2-MCQ] [2 marks]

Brass is:

- (A) Cu-Zn
 - (B) Cu-Sn
 - (C) Cu-Ni
 - (D) Cu-Al
-

Q53. [2-MCQ] [2 marks]

Brass is:

- (A) Cu-Zn
 - (B) Cu-Sn
 - (C) Cu-Ni
 - (D) Cu-Al
-

Q54. [1-MCQ] [1 mark]

Reducing agent in blast furnace:

- (A) Coke
 - (B) Limestone
 - (C) Hot air
 - (D) Slag
-

Q55. [1-MCQ] [1 mark]

Blast furnace main product:

- (A) Steel
 - (B) Pig iron
 - (C) Copper
 - (D) Aluminum
-

Q56. [1-MCQ] [1 mark]

Halls-Heroult process produces:

- (A) Iron
 - (B) Aluminum
 - (C) Copper
 - (D) Zinc
-

Q57. [2-MCQ] [2 marks]

Halls-Heroult process produces:

- (A) Iron
 - (B) Aluminum
 - (C) Copper
 - (D) Zinc
-

Q58. [2-MCQ] [2 marks]

Halls-Heroult process produces:

- (A) Iron
- (B) Aluminum
- (C) Copper

(D) Zinc

Q59. [2-MCQ] [2 marks]

Blast furnace main product:

- (A) Steel
- (B) Pig iron
- (C) Copper
- (D) Aluminum

Q60. [1-MCQ] [1 mark]

Edge dislocation moves by:

- (A) Climb
- (B) Glide
- (C) Cross slip
- (D) None

Q61. [1-MCQ] [1 mark]

Eutectic point has:

- (A) One phase
- (B) Two phases
- (C) Three phases simultaneously
- (D) Four phases

Q62. [1-NAT] [1 mark]

Martensite is formed by:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-NAT] [1 mark]

Cathodic reaction in rusting:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-MCQ] [1 mark]

Shielded metal arc uses:

- (A) Gas shield
- (B) Flux coated electrode
- (C) Both
- (D) None

Q65. [1-NAT] [1 mark]

Duralumin is Al +:

(Numerical Answer Type — Enter the numerical value)

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	A	1
3	1MCQ	C	1
4	1MCQ	B	1
5	1MCQ	A	1
6	1MCQ	C	1
7	1NAT	B	1
8	1NAT	B	1
9	1MCQ	B	1
10	1NAT	A	1
11	1MCQ	A	1
12	1MCQ	B	1
13	1MCQ	A	1

14	1NAT	C	1
15	1NAT	B	1
16	1MCQ	B	1
17	1MCQ	B	1
18	1MCQ	B	1
19	1MCQ	B	1
20	1NAT	B	1
21	1MCQ	A	1
22	1NAT	B	1
23	1MCQ	C	1
24	1MCQ	A	1
25	1MCQ	C	1
26	1MCQ	A	1
27	1NAT	A	1
28	1MCQ	C	1

29	1MCQ	A	1
30	1MCQ	A	1
31	2MCQ	A	2
32	2MCQ	A	2
33	2NAT	C	2
34	2MCQ	A	2
35	2MCQ	B	2
36	2MSQ	B	2
37	2MCQ	B	2
38	2MSQ	A	2
39	2MCQ	A	2
40	2MCQ	B	2
41	2MCQ	C	2
42	2NAT	A	2
43	2MCQ	A	2

44	2MCQ	A	2
45	2MCQ	B	2
46	2MCQ	A	2
47	2NAT	A	2
48	2MCQ	C	2
49	2MCQ	B	2
50	2MCQ	B	2
51	2MCQ	C	2
52	2MCQ	A	2
53	2MCQ	B	2
54	2NAT	B	2
55	2MCQ	A	2
56	2NAT	B	2
57	2MCQ	A	2
58	2MCQ	B	2

59	2MCQ	B	2
60	2MCQ	C	2
61	2MCQ	A	2
62	2MCQ	B	2
63	2MCQ	A	2
64	2MSQ	B	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: difficult

Q3.

Answer: C

Solution:

Correct Answer: C

Subject: Physical Metallurgy | Topic: Crystal Structure | Difficulty: moderate

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Mechanical Metallurgy | Topic: Dislocations | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Extractive Metallurgy | Topic: Iron Making | Difficulty: difficult

Q6.

Answer: C

Solution:

Correct Answer: C

Subject: Phase Transformations | Topic: Phase Diagrams | Difficulty: moderate

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Heat Treatment | Topic: Steel | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Corrosion | Topic: Electrochemical | Difficulty: easy

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Welding & Joining | Topic: Arc Welding | Difficulty: difficult

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Non-Ferrous Metals & Alloys | Topic: Aluminum | Difficulty: difficult

Q11.

Answer: A

Solution:

Eutectic: L $\rightarrow$ alpha + beta at eutectic temperature. One liquid to two solids.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate

Q12.

Answer: B

Solution:

BCC: 8 nearest neighbors at cube corners.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy

Q13.

Answer: A

Solution:

Brass: copper-zinc alloy. Bronze: copper-tin.

Subject: Material Science | Topic: Material Science | Difficulty: moderate

Q14.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q15.

Answer: B

Solution:

BCC: 8 nearest neighbors. Coordination number = 8.

Subject: Material Science | Topic: Material Science | Difficulty: moderate

Q16.

Answer: B

Solution:

$(1 + 1/n)^{2n} = [(1 + 1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q17.

Answer: B

Solution:

Blast furnace: iron ore + coke + limestone $\rightarrow$ pig iron (high carbon iron).

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate

Q18.

Answer: B

Solution:

Hall-Heroult: electrolytic reduction of alumina (Al_2O_3) to aluminum.

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate

Q19.

Answer: B

Solution:

Lever rule: fraction of phase = opposite arm / total tie-line length.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy

Q20.

Answer: B

Solution:

Quenching: rapid cooling from austenitizing temp. Forms hard martensite.

Subject: Material Science | Topic: Material Science | Difficulty: moderate

Q21.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4}(1 + 4) = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q22.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q23.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q24.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q25.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q26.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: C

Solution:

FCC: 12 nearest neighbors. Coordination number = 12.

Subject: Material Science | Topic: Material Science | Difficulty: easy

Q29.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q31.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q33.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q38.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q40.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q41.

Answer: C

Solution:

Rolle's: f continuous on $[a, b]$, differentiable on (a, b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a, b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q43.

Answer: A

Solution:

$L\{t^n f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q44.

Answer: A

Solution:

$$f(i) = i^2 + 3i + 2 = -1 + 3i. \operatorname{Re}(f(i)) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q45.

Answer: B

Solution:

Quenching: rapid cooling from austenitizing temp. Forms hard martensite.

Subject: Material Science | Topic: Material Science | Difficulty: difficult

Q46.

Answer: A

Solution:

Eutectic: L $\rightarrow$ alpha + beta at eutectic temperature. One liquid to two solids.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy

Q47.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q49.

Answer: B

Solution:

BCC: 8 nearest neighbors at cube corners.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy

Q50.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q51.

Answer: C

Solution:

FCC: 12 nearest neighbors. Coordination number = 12.

Subject: Material Science | Topic: Material Science | Difficulty: easy

Q52.

Answer: A

Solution:

Eutectic: L $\rightarrow$ alpha + beta at eutectic temperature. One liquid to two solids.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy

Q53.

Answer: B

Solution:

Hall-Heroult: electrolytic reduction of alumina (Al_2O_3) to aluminum.

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: easy

Q54.

Answer: B

Solution:

BCC: 8 nearest neighbors. Coordination number = 8.

Subject: Material Science | Topic: Material Science | Difficulty: moderate

Q55.

Answer: A

Solution:

Brass: copper-zinc alloy. Bronze: copper-tin.

Subject: Material Science | Topic: Material Science | Difficulty: difficult

Q56.

Answer: B

Solution:

BCC: 8 nearest neighbors at cube corners.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate

Q57.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q58.

Answer: B

Solution:

Lever rule: fraction of phase = opposite arm / total tie-line length.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy

Q59.

Answer: B

Solution:

Hall-Heroult: electrolytic reduction of alumina (Al_2O_3) to aluminum.

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: easy

Q60.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q61.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q62.

Answer: B

Solution:

Blast furnace: iron ore + coke + limestone $\rightarrow$ pig iron (high carbon iron).

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: easy

Q63.

Answer: A

Solution:

Brass: copper-zinc alloy. Bronze: copper-tin.

Subject: Material Science | Topic: Material Science | Difficulty: difficult

Q64.

Answer: B

Solution:

Lever rule: fraction of phase = opposite arm / total tie-line length.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate

Q65.

Answer: B

Solution:

Lever rule: fraction of phase = opposite arm / total tie-line length.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: difficult

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 32 questions, 52 marks
Physical Metallurgy: 11 questions, 18 marks
Material Science: 9 questions, 14 marks
Extractive Metallurgy: 6 questions, 9 marks
General Aptitude: 1 questions, 1 marks
Mechanical Metallurgy: 1 questions, 1 marks
Phase Transformations: 1 questions, 1 marks
Heat Treatment: 1 questions, 1 marks
Corrosion: 1 questions, 1 marks
Welding & Joining: 1 questions, 1 marks
Non-Ferrous Metals & Alloys: 1 questions, 1 marks

Difficulty Distribution:

Easy: 21 | Moderate: 31 | Difficult: 13

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.